NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

Department of Computer Science & Engineering

Mid Semester Examination, Autumn 2023 Duration: 2 Hours

Subject: Machine Learning Subject Code: CS6218 Full Marks: 30

Answer All questions. All parts of a question must be answered at one place. Mere answer without justification will not fetch any mark.

1. (a) Consider the following generative model for a 2-class classification problem, in which the class conditionals are Bernoulli distributions: $p(\omega_1) = \pi$, $p(\omega_2) = 1 - \pi$. [3]

$$x|\omega_1 = \begin{cases} 1 \text{ with probability } 0.5, \\ 0 \text{ with probability } 0.5 \end{cases} \qquad x|\omega_2 = \begin{cases} 1 \text{ with probability } 0.5, \\ 0 \text{ with probability } 0.5 \end{cases}$$

Assume the loss matrix

$$\begin{array}{rc} & \begin{array}{cc} \text{true class=1} & \text{true class=2} \end{array} \\ \begin{array}{r} \text{predicted class=1} \\ \text{predicted class=0} \end{array} & \begin{bmatrix} 0 & \lambda_{12} \\ \lambda_{21} & 0 \end{bmatrix} \end{array}$$

Give a condition in terms of λ_{12}, λ_{21}, and π that determines when class 1 should always be chosen as the minimum-risk class.

(b) Let's derive the decision boundary when one class is Gaussian and the other class is exponential. Our feature space is one-dimensional ($d = 1$), so the decision boundary is a small set of points. We have two classes, named N for normal and E for exponential. For the former class ($Y = N$), the prior probability is $P(Y = N) = \frac{\sqrt{2\pi}}{\sqrt{1+2\pi}}$ and the class conditional $P(X|Y = N)$ has the normal distribution $N(0, \sigma^2)$. For the latter, the prior probability is $P(Y = E) = \frac{1}{\sqrt{1+2\pi}}$ and the class conditional has the exponential distribution [3]

$$P(X = x|Y = E) = \begin{cases} \lambda e^{-\lambda x} \text{ if } x \geq 0; \\ 0 \text{ otherwise.} \end{cases}$$

Write an equation in x for the decision boundary. (Only the positive solutions of your equation will be relevant; ignore all $x < 0$.) Use the 0-1 loss function. Simplify the equation until it is quadratic in x.

(c) Find the illness class predicted by the Naive Bayes Classifier for the input vector $X = (Chennai, Female)$ given the following table. [4]

city	gender	illness
Bangalore	Male	Yes
Chennai	Female	Yes
Chennai	Male	Yes
Chennai	Female	Yes
Bangalore	Female	Yes
Bangalore	Male	No
Chennai	Female	No
Chennai	Male	No

2. (a) Define Information Gain. Construct a decision tree based on the following data taking information gain as the splitting criterion. [7]

salary	city	job	sex	class
100	Bangalore	employed	male	yes
90	Chennai	student	male	yes
80	Chennai	employed	male	no
150	Chennai	unemployed	male	no
120	Bangalore	employed	female	yes
30	Bangalore	student	female	no
60	Chennai	employed	female	yes

(b) What is gini impurity index? Compute the gini impurity index for the attribute city in the previous example. [2]

(c) How to deal with overfitting in Decision tree learning. [1]

3. (a) Build a linear regression model to find the regression parameters for the following dataset. [4]
Do not use the Normal Equation. Use at least two iterations of gradient descent.

x_1	x_2	y
0	1	6
0	0	4
1	0	5
1	1	7

(b) Show how the Generalization error (Expected Test error) can be decomposed into Bias, Variance and Noise. Draw the Bias-Variance vs model complexity graph. Suppose you have high Bias and low Variance: what measures should be taken to mitigate the problem? [3]

(c) Prove that sum $z = x + y$ of two independent random variables x, and y where $x \sim N(\mu_x, \sigma_x^2)$ and $y \sim N(\mu_y, \sigma_y^2)$ is also Gaussian with mean value and variance equal to $\mu_x + \mu_y$, and $\sigma_x^2 + \sigma_y^2$. [3]

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